

Usability Testing: Exercises

Human Computer Interaction

Luigi De Russis

Academic Year 2025/2026

Goal

- Recognize good/bad tasks in a usability test plan
 - and know how to fix/improve them
- Draft a script for a sample usability test

Recap: Usability Testing

- Goals
 - Identify problems in the design
 - Uncovering opportunities to improve
 - Learning about the target user's behavior and preferences
- Definition (by the Nielsen-Norman Group)
 - In a usability-testing session, a “*facilitator*” (also called, “moderator”) asks a *participant* to perform *tasks*, usually using one or more specific user interfaces.
 - While the participant completes each task, the researcher observes the participant's behavior and listens for feedback.

Recap: Phases of Usability Testing

1. Plan

- who are your participants? what are you going to test, where, and how?

2. Run

- one participant at time, multiple sessions
- collect data about the interactive system/interface

3. Analyze

- extract information from the collected data, both qualitative and quantitative

Recap: Tasks

- Define tasks according to the main goals that target users may have on your system
- Rather than simply ordering participants to “do X” with no explanation, it is better to situate the request within a short **scenario**:
 - it sets the stage for the action and explains why the user is “doing X”
- A task scenarios for usability testing need to provide **context** so users engage with the interface and pretend to perform it as if they were at home or in the office

<https://www.nngroup.com/articles/task-scenarios-usability-testing/>

Recap: Tasks

- Make the task **realistic**
- Make the task **actionable**
- Avoid giving **clues** and **describing the steps**

Exercise 1

- Look at the task list extracted from a plan of an **imaginary** usability testing, available in the next slide
- Answer those **questions**:
 - Are tasks good? Why/why not?
 - Are success criteria appropriate? Why/why not?
 - Is the listed methodology appropriate: in general, for the task, for the collected metrics?
- **Duration**: ~ 15 minutes
- In-class discussion will follow

Exercise 1: Task List

#	Task	Success Criteria	Methodology	Metrics
1	The participants login via the homepage	The login form appears	Think-aloud	Time on task
2	The participants logout from the homepage after login	The logout button appears	Think-aloud	Time on task
3	Check the detailed information of a garment	The garment page appears	Think-aloud	-
4	Participant clicks the button to contact the shop from the garment info page	The message section appears	Think-aloud	-
5	Participant sends a message to the shop	Message sent	-	-
6	Participant goes back to the homepage	Click the header logo	-	-

Exercise 2

- Look at the task list extracted from a plan of a usability testing made by “Team N” in the 2023 version of the course, available in the next two slides
- **Topic:** help non-STEM university students prepare computer science exams
- Answer those **questions**:
 - Are tasks good? Why/why not?
 - Are success criteria appropriate? Why/why not?
 - Is the listed methodology appropriate: in general, for the task, for the collected metrics?
- **Duration:** ~ 30 minutes
- In-class discussion will follow

Exercise 2: Task List (I)

Task-related metrics: time on tasks, success rate, number of errors

#	Task	Success Criteria	Methodology
1	Select the “Web Development basics” topic in the topics section after searching for the word “Development”	The participant executes a search for the word “Development” and selects the correct topic	Think-aloud
2	Do a pending exercise after filtering for completion with the value “in progress”	The participant correctly identifies and selects the pending exercise from the filtered list	Think-aloud
3	Start an incomplete exercise on a topic where your progress is 3/5 and sends it without writing anything	The participant successfully navigates to the specified incomplete exercise on the topic with a progress rating of 3/5 and sends it without adding any written content	Think-aloud

Exercise 2: Task List (II)

Task-related metrics: time on tasks, success rate, number of errors

#	Task	Success Criteria	Methodology
4	Do the only “in progress” exercise of Mobile Application Development to complete all the topic’s exercises	The participant provides functional code that solves the proposed problem and on the page that displays the list of exercises of the “Mobile App Development” topic, a message is shown indicating that all exercises have been completed	Think-aloud
5	Propose an exercise for a completed topic	The user-created exercise is added to the 'My Exercises' list	Think-aloud
6	Sort the proposed exercises by descendant name	The participant accurately applies the “descendant name” sorting option	Think-aloud
7	Update a rejected exercise proposal	The participant revises the rejected exercise proposal in accordance with provided feedback or identified issues	Think-aloud

Exercise 3

- Start to draft a sample plan and script for conducting the usability testing of the **PoliTo Student App**
- Make the needed decisions (participants, equipment, artifacts, tasks, ...)
 - We will make some of those together
- Work using the following Google Doc as a template:
<https://docs.google.com/document/d/1cJNp86wt8sjDkjin2BphpcRBZpY8lvQoDDlrycOJP92k>
- In-class discussion will follow

License

- These slides are distributed under a Creative Commons license “**Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0)**”
- **You are free to:**
 - **Share** — copy and redistribute the material in any medium or format
 - **Adapt** — remix, transform, and build upon the material
 - The licensor cannot revoke these freedoms as long as you follow the license terms.
- **Under the following terms:**
 - **Attribution** — You must give [appropriate credit](#), provide a link to the license, and [indicate if changes were made](#). You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
 - **NonCommercial** — You may not use the material for [commercial purposes](#).
 - **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the [same license](#) as the original.
 - **No additional restrictions** — You may not apply legal terms or [technological measures](#) that legally restrict others from doing anything the license permits.
- <https://creativecommons.org/licenses/by-nc-sa/4.0/>

